IFN647 Week 7 Review Questions

Question 1 (Mutual Information)

For two words (or terms) *a* and *b*, their mutual information measure (*MIM*) is defined as

$$\log \frac{P(a,b)}{P(a)P(b)}$$

where $P(a)$ is the probability that word *a* occurs in a text window of a given size, $P(b)$ is the probability that word *b* occurs in a text window, and $P(a, b)$ is the probability that *a* and *b* occur in the same text window.

Assume a function *df*(*docs*, *a*) is defined to calculate the number of documents containing word *a*, and function *df*2(*docs*, *a*, *b*) returns the number of documents containing both words *a* and *b*, where *docs* is a set of documents.

(1) Let *D* be a set of documents. For any two words *a* and *b*, use the *MIM* equation and the provided two functions to calculate their *MIM*.

(2) Assume the collection of documents is represented as a dictionary {*docID*: {*term*: *counts*, …},…}, for example, *docs* = {'D1':{'term1':3, 'term4':5, 'term5':7},'D2':{'term1':5, 'term2':3, 'term3':4, 'term4':6}, 'D3':{'term3':5, 'term4':4, 'term5':6}, 'D4':{'term1':9, 'term4':1, 'term5':2}, 'D5':{'term2':1, 'term4':3, 'term5':2},'D6':{'term1':3, 'term3':2, 'term4':4, 'term5':4}}.
Define a python function *df*2(*docs*, *a*, *b*) to calculate n_{ab} - the number of documents that include both terms *a* and *b*.

(3) Assume python function *c_df*(*docs*) calculates *df* value for all terms in *docs* and returns a {*term*:*df*, …} dictionary. Use both functions *df*2() and *c_df*() to define a Python function *MIM*(*docs*, *a*, *b*) to calculate the *MIM* value for terms *a* and *b*.

Question 2 (Term frequency measure)

Assume Table 1 is a training set provided by user feedback. A term's frequency measure is defined as follows:

$$tf(t_k) = \frac{\sum_{d_i \in D^+} f_{ik}}{\sum_{d_i \in D^+} \sum_{t_j \in T} f_{ij}}$$

Doc_ID	**Terms**	**Label (Relevant)**
D_1	GERMAN, VW	0 no
D_2	US, US, ECONOM, SPY	1 yes
D_3	US, BILL, ECONOM, ESPIONAG	1 yes
D_4	US, ECONOM, ESPIONAG, BILL	1 yes
D_5	GERMAN, MAN, VW, ESPIONAG	0 no
D_6	GERMAN, GERMAN, MAN, VW, SPY	0 no
D_7	US, MAN, VW	0 no

Table 1. A training set from user feedback

Calculate term ESPIONAG's frequency measure, and show your calculation steps:

Question 3 (<u>Probabilistic information filtering model)</u>

For a given training set *D* (or labelled dataset, e.g., a list of lists of terms, where each list of terms denotes a document), which consists of a set of relevant documents *Rel* and a set of non-relevant documents *Nonrel*; BM25 based IF model uses the following equation to calculate a term's weight:

$$w_5(t) = \log \frac{\dfrac{r(t)+0.5}{R-r(t)+0.5}}{\dfrac{n(t)-r(t)+0.5}{(N-n(t))-(R-r(t))+0.5}}$$

Provide equations to calculate N, R, $r(t)$ and $n(t)$, where an equation is a math formula or a python definition.

<u>Question 4</u> Which of the following descriptions is wrong? And justify your answer.

(1) Large language models (LLMs) can naturally generate human-readable explanations for recommendations, thereby improving system transparency and interpretability.

(2) LLM-based information filtering does not require labelled training data, as well-designed prompts can effectively guide the model's behaviour.
(3) Retrieval-Augmented Generation (RAG) is an AI framework that enhances LLMs by retrieving relevant information from external knowledge bases, leading to more accurate and factually grounded outputs.
(4) LLMs such as GPT-4 have demonstrated strong capabilities in natural language understanding, and their effectiveness in real-world news recommendation systems has been fully established.
(5) Chain-of-Thought (CoT) prompting is a method to improve LLMs by prompting them to generate step-by-step reasoning before giving a final answer.